

Beyond BioID: Streptavidin outcompetes antibody fluorescence signals in protein localization and readily visualises targets evading immunofluorescence detection

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Abstract

Immunofluorescence is a common method to localise proteins within their cellular context via fluorophore labelled antibodies and for some applications without alternative. However, some protein targets evade detection due to low protein abundance or accessibility issues. In addition, some imaging methods require a massive reduction in antigen density thus impeding detection of even medium-abundant proteins.

Here, we show that the fusion of the target protein to TurboID, a biotin ligase labelling lysine residues in close proximity, and subsequent detection of biotinylation by fluorescent streptavidin offers an “all in one” solution to the above-mentioned restrictions. For a wide range of target proteins tested, the streptavidin signal was significantly stronger than an antibody signal, markedly improving the imaging sensitivity in expansion microscopy and correlative light and electron microscopy, with no loss in resolution. Importantly, proteins within phase-separated regions, such as the central channel of the nuclear pores, the nucleolus or RNA granules, were readily detected with streptavidin, while most antibodies fail to label proteins in these environments. When TurboID is used in tandem with an HA epitope tag, co-probing with streptavidin and anti-HA can be used to map antibody-accessibility to certain cellular regions. As a proof of principle, we mapped antibody access to all trypanosome nuclear pore proteins (NUPs) and found restricted antibody labelling of all FG NUPs of the central channel that are known to be phase-separated, while most non-FG Nups could be labelled. Lastly, we show that streptavidin imaging can resolve dynamic, temporally and spatially distinct sub-complexes and, in specific cases, reveal a history of dynamic protein interaction.

In conclusion, streptavidin imaging has major advantages for the detection of lowly abundant or inaccessible proteins and in addition, can provide information on protein interactions and biophysical environment.

eLife assessment

This **valuable** study demonstrates how proximity labeling with streptavidin can be used to boost fluorescence signals in otherwise hard-to-label regions of cells. The experimental verification of amplification of fluorescence near epitope tags in phase-separated compartments is **solid**, demonstrating enhanced signal-to-noise compared to immunofluorescence. This study will be of particular interest to those using correlative light and electron microscopy or expansion microscopy when the signal is limiting or inaccessible.

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Introduction

Visualisation of a protein within the cellular context is routinely achieved either by fusing it to a fluorescent protein, or by using fluorophore labelled antibodies specific for a target protein or peptide tag that is genetically fused to it. Fluorescent proteins are not suitable for all imaging applications, because fluorescence is lost under fixation and permeabilization conditions. Applications that are fully dependent on antibodies include correlative electron and light microscopy (CLEM) applications, expansion microscopy or protein detection in combination with *in situ* hybridisation for nucleic acid detection.

Sometimes, antibodies fail to detect a protein at its expected localisation, which is rarely followed up or reported. We recently observed the absence of an anti-HA antibody signal at the nuclear pores for endogenously expressed trypanosome mRNA export factor MEX67 fused to the small peptide hemagglutinin (HA) epitope tag: the anti-HA signal was entirely restricted to the nucleoplasm (Moreira et al., 2023). Likewise, a polyclonal antibody raised against *T. brucei* MEX67 did not stain the nuclear pores (Pozzi et al., 2023). In contrast, fusions of *T. brucei* MEX67 to fluorescent proteins (either N- or C-terminally) are primarily detected at the nuclear pores with some signal extending to the nucleoplasm (Billington et al., 2023; Kramer et al., 2010). This is in agreement with the reported function of *T. brucei* MEX67 in mRNA export (Dostalova et al., 2013; Schwede et al., 2009) and with nuclear pore localisation of MEX67 orthologues in other eukaryotes (Katahira et al., 1999; Köhler and Hurt, 2007; Mangus et al., 2003; Rodriguez et al., 2004; Stewart, 2019; Sträßer et al., 2000; Terry and Wente, 2007) where Mex67 was even termed a mobile nucleoporin (Derrer et al., 2019).

We next expressed *Tb*MEX67 fused to a TurboID-HA tandem tag. TurboID is a biotin ligase, widely employed for proximity labelling techniques facilitating analysis of protein interactions *in vivo* (Branon et al., 2018). To our surprise, fluorophore labelled streptavidin readily delivered a signal localizing primarily to the nuclear pores and partially to the nucleoplasm, consistent with the localisation of MEX67 fused to fluorescent proteins (Moreira et al., 2023). We therefore concluded that the streptavidin signal shows the correct localization of MEX67, while antibodies fail to bind MEX67 at the nuclear pore and selectively stain the nucleoplasmic fraction of MEX67. One possible reason for the absence of antibody stain could be the localisation of MEX67 to the central channel of the nuclear pore that is lined by phenylalanine-glycine nucleoporins (FG NUPs), intrinsically disordered proteins dominated by FG-repeats, that assemble to a dense meshwork and create a phase-separated environment, physically distinct from the rest of the cell (Davis et al., 2022; Nag et al., 2022). It remains unclear, why streptavidin was able to stain biotinylated proteins within these antibody inaccessible regions, but possible reasons are: (i) tetrameric streptavidin is smaller and more compact than IgGs (60 kDa versus a tandem of two IgGs, each

with 150 kDa) (ii) the interaction between streptavidin and biotin is ~100 fold stronger than a typical interaction between antibody and antigen and (iii) streptavidin contains four fluorophores, in contrast to only one per secondary IgG.

Here, we set out to investigate whether streptavidin imaging offers a general solution to antibody-accessibility problems. We expressed 11 trypanosome and one mammalian protein of different *bona fide* phase-separated regions (nucleolus, stress granules, central channel of the nuclear pores) fused to a TurboID-HA tandem tag. All proteins could be readily visualised with streptavidin, while they could not be labelled with anti-HA. Importantly, we noticed that streptavidin imaging has further major advantages to antibody labelling, even for antibody-accessible proteins: (i) the multiple biotinylation sites of the bait and adjacent proteins provided a massive boost in signal, which is in particular useful in applications with diluted antigens, such as expansion microscopy or correlative light and electron microscopy (CLEM); (ii) since biotinylation extends to adjacent proteins and is stable over the lifetime of a protein, protein interactions are preserved in time, enabling to monitor dynamic processes in the cell.

Materials and methods

Culture and genetic modification of trypanosomes

T. b. brucei Lister 427 procyclic cells were cultured in SDM-79 (Brun and Schönenberger, 1979). Generation of transgenic trypanosomes was performed using standard methods (McCulloch et al., 2004 [↗](#)). Almost all trypanosome fusion proteins were expressed from the endogenous locus, by modifying one allele via a PCR-based approach reliant on a (modified) version of the pPOT vector system (Dean et al., 2015 [↗](#)). Occasionally, a plasmid-based system was used for endogenous tagging (Kelly et al., 2007 [↗](#)). For overexpression of ALPH1-eYFP, a tetracycline inducible system was used (Sunter et al., 2012 [↗](#)) and expression was induced for 24 hours. Details about all fusion constructs are provided in Table S1. The TurboID-HA tandem tag encodes the TurboID open reading frame, followed by a two-amino acid long spacer (AS), followed by the HA tag (YPYDVDPYA). The TurboID-Ty1 tandem tag encodes the TurboID open reading frame, followed by a short linker (ASGSGS), followed by the Ty1 tag (EVHTNQDPLD).

Culture and genetic manipulation of HeLa cells

pEGFP-N1 was modified to contain the sequence encoding TurboID-4HA with a stop-codon upstream of the eGFP sequence. Sequences of the open reading frames of NUP88 and NUP54 were amplified from human cDNA and cloned in frame downstream of the TurboID-4HA sequence. This results in expression of NUP88 or NUP54 fused to the N-terminal TurboID-4HA tag, without the eGFP. The sequences of the fusion proteins are provided in Table S1. HeLa cells were grown in Dulbecco's modified Eagle's medium (DMEM; Gibco by Life Technologies, Darmstadt, Germany) supplemented with 10% fetal calf serum (FCS; Capricorn Scientific, Ebsdorfergrund, Germany) and 1% penicillin-streptomycin (Thermo Fisher Scientific, Dreieich, Germany) at 37°C and 5% CO₂. Cells were transfected with the respective TurboID-HA fusion constructs using Effectene™ (Qiagen, Hilden, Germany) following the manufacturer's protocol and incubated for ~24 hours in a petri-dish containing a coverslip. Immunofluorescence/streptavidin labelling was performed on cells attached to the coverslips, as detailed below for trypanosomes.

Immunofluorescence and streptavidin labelling

About 1×10^7 procyclic-form *T. brucei* cells, harvested at a density of 5×10^6 cells/ml, were washed once in 1 ml SDM79 without hemin and serum and resuspended in 500 µl PBS. For fixation, 500 µl of 8% paraformaldehyde was added for 20 min while rotating. After addition of 7 ml PBS supplemented with 20 mM glycine, cells were pelleted, resuspended in 150 µl PBS, and spread on poly-lysine-coated slides (within circles drawn with a hydrophobic pen). Cells were allowed to

settle for 15 min, before removing surplus PBS and permeabilizing cells with 0.5% Triton X-100 in PBS. Slides were rinsed in PBS, and cells were then blocked in 3% bovine serum albumin (BSA) in PBS for 30 min, followed by 60 min incubation with rabbit mAb-anti-HA C29F4 (1:500 dilution; Cell Signaling Technology) and Streptavidin-Cy3 (1:200 dilution; Jackson Laboratories) in PBS/3% BSA. Slides were washed in PBS (three times for 5 min), then incubated with anti-rabbit Alexa Fluor Plus 488 (1:500 dilution, A32731 Invitrogen) in PBS/3% BSA for 50 min and a further 10 min upon addition of 4',6-diamidino- 2-phenylindole (DAPI) (0.1 µg/ml). Slides were washed 3 × 5 min in PBS and embedded into ProLong Diamond Antifade Mountant (Thermo Fisher Scientific). For control experiments, the following antibodies/nanobodies were used: Ty1-tag monoclonal antibody BB2 (hybridoma supernatant, 1:300); *T. brucei* MEX67 polyclonal antiserum (1:1000, kind gift of Mark Carrington) and anti-GFP nanobody (GFP-Booster Alexa Fluor 647 Chromotek gb2AF647, 1:1000).

Microscopy

For most fluorescence microscopy experiments, images were acquired using a DMI8 widefield microscope (Thunder Imager, Leica Microsystems, Germany) with a HCX PL APO CS objective (100x, NA = 1.4, Leica Microsystems) and Type F Immersion Oil (refractive index = 1.518, Leica Microsystems). The microscope was controlled using LAS-X software (Leica Microsystems). Samples were illuminated with an LED8 light source (Leica). Excitation light was selected by using the following filter sets: Ex 391/32 nm; DC 415nm; Em 435/30 nm (DAPI), 436/28 nm; DC 459 nm; Em 519/25 nm (AlexaFluor™ 488), 554/24 nm; DC 572 nm; Em 594/32 nm (Cy3), 578/24 nm; DC 598 nm; Em 641/78 nm (AlexaFluor™ 594). Images were captured using a K5 sCMOS camera (Leica, 6.5 µm pixel size). Between 50 and 70 images in 140 nm distances were recorded for each Z-stack. Exposure times were 100-200 ms for DAPI and between 200-800 ms for all other fluorophores. For some images, an inverted wide-field microscope Leica DMI6000B (Leica Microsystems GmbH, Wetzlar, Germany) equipped with a 100x oil immersion objective (NA 1.4) and a Leica DFC365 camera (6.5 µm/pixel) was used. Filter sets were (i) ex: 340–380 nm, dc: 400nm, em: 450–490 nm (DAPI), (ii) ex: 450–490 nm, dc: 495nm, em: 500–550 nm (eYFP), (iii) ex: 530– 560nm, dc: 570 nm, em: 572- 648 nm (TMR). Image acquisition was done like on the thunder imager. Images are shown either unprocessed (raw data) or processed by instant computational clearing (Software provided by the Thunder imager) or deconvolution using Huygens Essential software (Scientific Volume Imaging BV). Presentations are either as single plane or as Z-stack projection (sum slices), as indicated. Scanning electron microscopy and correlation with light microscopy was done as previously described (Kramer et al., 2020 [DOI](#)).

Affinity enrichment of biotinylated proteins and on beads digestion

The isolation of biotinylated proteins from trypanosomes and preparation for mass spectrometry was done as previously described (Moreira et al., 2023), except that 1 mM biotin was added during the elution step from the streptavidin beads by tryptic digest, to improve recovery of biotinylated peptides.

Western blots

Western blotting was performed using standard methods. Detection of biotinylated proteins was done with Streptavidin-IRDye®680LT (LI-COR) (1:20,000). Proteins with HA-tag(s) were detected using rat mAb-anti-HA 3F10 (Roche, 1:1000) as primary antibody and IRDye® 800CW Goat anti-Rat IgG (Licor, 1:20,000) as secondary antibody.

Mass spectrometry and proteomics analysis

TurboID-eluted peptides were resuspended in 50 mM NH₄HCO₃ and passed over C18 stage tip columns as described (Rappsilber et al., 2007 [DOI](#)) and then analyzed by LC-MS/MS on an Ultimate3000 nano rapid separation LC system (Dionex) coupled to an LTQ Orbitrap Fusion mass

spectrometer (Thermo Fisher Scientific). Minimum peptide length was set at seven amino acids allowing a maximum of two missed cleavages, and FDRs of 0.01 were calculated at the levels of peptides, proteins, and modification sites based on the number of hits against the reversed sequence database. Spectra were processed using the intensity-based label-free quantification (LFQ) in MaxQuant version 2.1.3.0 (Cox and Mann, 2008 [DOI](#); Cox et al., 2014 [DOI](#)) searching the *T. brucei* strain TREU927 annotated protein database (version 63) from TriTrypDB (Aslett et al., 2010 [DOI](#)). Statistical analyses of LFQ data were done in Perseus (Tyanova et al., 2016 [DOI](#)). LFQ values were filtered for at least one valid value in the bait sample group, log2 transformed and missing values imputed from a normal distribution of intensities around the detection limit of the mass spectrometer. These values were subjected to a Student's t-test comparing the respective bait sample groups to an untagged control (wt parental cells) triplicate sample group. The resulting *t*-test difference is the difference between the means of logarithmic abundances of a protein group. Bait samples were prepared in duplicate (NUP76, NUP96, NUP110) or triplicate (MEX67, NUP158). All proteomics data have been deposited at the ProteomeXchange Consortium via the PRIDE partner repository (Perez-Riverol et al., 2019 [DOI](#)) with the dataset identifier PXD031245 (MEX67, NUP158, wt control, GFP control) and PXD047268 (NUP76, NUP96, NUP110).

Ultrastructural Expansion Microscopy (U-ExM) with procyclic *T. brucei*

The U-ExM method was performed as described in (Gambarotto et al., 2020 [DOI](#)) with some modifications. 1×10^7 trypanosome cells were harvested (1500 g, 10 min) and the pellet resuspended in 500 μ l PBS. 500 μ l 8% formaldehyde (FA, 36.5-38%, F8775, Sigma)/8% acrylamide (AA, 40%, A4058, Sigma) diluted in PBS was added to achieve a final concentration of 4%FA/4%AA. Cells were allowed to settle on a 12 mm round coverslip treated with poly-L-lysine (P8920, Sigma) placed in a well of a 24-well plate. The adjacent wells were filled with ddH₂O, the plate sealed with parafilm and incubated at 37°C for 5 hours. A small piece of parafilm was placed on top of a humid chamber, where a drop of monomer solution (sodium acrylate 23% (w/v) (408220, Sigma), AA 10% (w/v), N-N'-methylenebisacrylamide 0.1% (w/v) (M7279, Sigma) in PBS) was added. The coverslip was placed on top of the drop, with the cells facing down. The coverslip was removed, cleaned from excess liquid with a laboratory wipe and placed in 40 μ l of monomer solution freshly complemented with 1 μ l 10% APS (327085000, Acros Organic) and 1 μ l 10% TEMED (T7024, Sigma). Sodium acrylate stock solution 38% (w/v) was prepared by dissolving 1.9 g sodium acrylate in 3.1 ml of ddH₂O followed by filtering through 0.2 μ m filter. The humid chamber was closed, and the gel allowed to polymerize for one hour at 37°C. Coverslips were transferred to a well of a 6-well plate containing 1 ml denaturation buffer (200 mM SDS (A1112, Applichem), 200 mM NaCl (A2942, Applichem) and Tris-HCl pH 9 (5429.3, Roth)) and incubated for 15 minutes with gentle shaking. Gels were detached from the coverslips with a spatula and transferred to a 1.5 mL Eppendorf tube. The tube was filled with denaturation buffer to the top, closed and incubated at 95°C for 90 minutes in a heating block. Gels were transferred to a beaker with 100 ml ddH₂O for the first round of expansion for 30 minutes. Water was exchanged and two more rounds of expansion were done, the last one being overnight at room temperature. The next day, gels were carefully transferred to a Petri dish and imaged to measure the expansion factor. Gels were transferred to a fresh beaker containing 100 ml PBS and incubated for 15 minutes. This was repeated once and then the gel was trimmed to a square of about 12 x 12 mm which was transferred to a 24-well plate for antibody incubation. Gels were incubated with 0.5 ml PBS BSA 3% (A1391, Applichem) with primary antibodies (mouse-anti-HA monoclonal antibodies (C29F4, Cell Signaling) at 1:500) at 37°C for three hours in a plate shaker at 1500 RPM. Gels were transferred to a 6-well plate for washing. They were washed three times with 10 ml PBS-T (0.1% Tween20 in PBS) for 10 minutes with agitation at room temperature. Gels were transferred back to a 24-well plate and incubated with 0.5 ml PBS BSA 3% with secondary antibodies AlexaFluor™ 488 goat anti-rabbit (A11008, Sigma, 1:500), Cy3-streptavidin and 0.1 μ g/ml DAPI (D9542, Sigma) at 37°C for three hours in a plate shaker at 1500 RPM. Gels were transferred to the 6-well plate and washed four times with 10 ml PBS-T for 10 minutes with agitation at room temperature, all steps from now on protected from

light. Gel pieces were expanded again in 100 ml ddH₂O in a beaker, with three water exchanges each 30 minutes and the last expansion step done overnight at room temperature. An imaging chamber Nunc™ Lab-Tek™ II with one chamber (734-2055, Thermofisher Scientific) was covered with a layer of poly-D-lysine (A3890401, Gibco) for one hour and left to dry overnight. For imaging, the gel pieces were trimmed to fit the chamber, excess water was removed with a laboratory wipe and the gel was placed in the imaging chamber, taking care not to shift after the initial placement. A drop of water was added to cover the top of the gel. The gel chamber was closed with the lid to avoid evaporation during imaging. For longer imaging sessions, drops of water were periodically added to prevent gel drying. Since cells can be on either side of the gel, the gel was divided into two pieces, one facing up and one facing down, to ensure cells can be detected, on either piece.

Protein retention Expansion Microscopy (proExM) in procyclic *T. brucei*

The proExM method was performed as described in (Tillberg et al., 2016 [DOI](#)) with some modifications. 5×10^7 cells were harvested by centrifugation at 1500 g for 10 minutes, washed once in 10 ml PBS and finally resuspended in 1 ml of PBS. 1 ml of PFA 8% (P6148, Sigma) was added and cells were fixed for 20 minutes at room temperature with rotation. 13 ml 20 mM glycine (50046, Sigma) in PBS (freshly prepared from a 2M stock) was added to quench and cells were harvested at 1500 g for 10 minutes. Cells were resuspended in 1 ml PBS and let to adhere to a 12 mm coverslip, previously treated with poly-L-lysine, placed in a well of a 24 well plate for 15 minutes. The well was rinsed with PBS and the coverslip incubated with PBS 0.5% Triton for 5 minutes for permeabilization. Wells were rinsed again with PBS and coverslips incubated with PBS 3% BSA for 30 minutes for blocking. 50 µl of primary antibody solution (rabbit anti-HA monoclonal antibodies (C29F4, Cell Signaling) 1:50 diluted in PBS BSA 1%) were placed on a piece of parafilm in a humid chamber and the cover slip was placed on top (cells facing down) for one hour in a humid chamber. The coverslip was returned to the well of the 24 well plate, cells facing up, and wells were washed with PBS (3 x 5 min) with agitation. 250 µl secondary antibodies (streptavidin-Cy3 (SA1010, Sigma) 1:1000; AlexaFluor™ 488 goat anti-rabbit (A11008, Sigma) 1:250) and DAPI ((D9542, Sigma) 1:1000) were added, diluted in 250 µl PBS BSA 1%. Wells were washed with PBS (4 x 5 min). From now on, all steps were done protected from light. To anchor the proteins to the gel, we prepared a solution of 5 mg AcX (Acryloyl-X, SE, 6-((acryloyl)amino)hexanoic Acid, Succinimidyl Ester, A20770, Invitrogen) in 500 µl anhydrous DMSO (D12345, Invitrogen); AcX solution was opened and aliquoted in a desiccated environment at – 20°C. AcX was diluted 1:100 in PBS, a drop was placed on a piece of parafilm in a humid chamber, the coverslips were placed top-down into the AcX drop and incubated at room temperature for 16 hours. Coverslips were returned to a well of a 24 well plate, washed twice for 5 minutes with PBS and stored in the fridge until the end of the day.

Monomer solution was prepared with sodium acrylate 8.6% (w/v) (408220, Sigma), AA 2.5% (w/v), N-N'-methylenebisacrylamide 1.5% (w/v) (M7279, Sigma) and NaCl 11.7% (w/v) (AM9760G, Ambion) in PBS; sodium acrylate stock solution 33% (w/v) was freshly prepared by dissolving 1.9 g sodium acrylate in 5 ml of ddH₂O, 0.2 µm-filtered. For gelation, gelling solution was prepared from 100 µl monomer solution, 1 µl 10% APS and 1 µl 10% TEMED, vortexed and used instantly. 50 µl of gelling solution was placed on a piece of parafilm in a humid chamber; the coverslips were placed on top, cells facing downwards, and incubated at 37°C for one hour. After polymerization, coverslips with the gel attached were transferred to a well of a 6-well plate with 1 ml of digestion solution (0.5% Triton X100, 1 mM EDTA pH 8 (AM9260G, Ambion), 50 mM Tris pH 8 (AM9855G, Ambion) and 1 M NaCl (AM9760G, Ambion) freshly supplemented with 10 µl proteinase K (P8107S, New England Biolabs). Plates were stored slightly tilted to ensure the coverslip was completely covered with digestion solution and incubated at room temperature overnight. Next day, gels were either already detached from the coverslip or could be easily detached with a spatula. Gels were

transferred to a Petri dish and three rounds of expansion, each with 20 ml ddH₂O for 20 minutes, were done. After this, a picture was taken to measure the expansion factor. Gel mounting in the imaging chamber and imaging were done as described for UExM.

Results

1. TurboID-fusion proteins can be accurately localised with fluorescent streptavidin by light microscopy

To establish a robust imaging system for TurboID fusion proteins by streptavidin, we expressed the *T. brucei* nuclear pore proteins NUP110, NUP76 and NUP96 fused to a TurboID-HA tandem tag in procyclic *Trypanosoma brucei* cells using both N and C-terminal fusion for each respective target protein. The tag is composed of the biotin ligase TurboID (Branon et al., 2018 [DOI](#)) and the HA peptide tag. Expression was constitutive from the endogenous locus and labelling relied on the biotin substate present in the medium (0.8 μ M in SDM79) (Moreira et al., 2023). Biotinylated proteins were purified by streptavidin affinity and analysed by liquid chromatography coupled to tandem mass spectrometry (LC-MS/MS), using both, wild type cells and cells expressing TurboID-GFP for background correction (Moreira et al., 2023). For each bait protein, we quantified and visualised the biotinylation of all other nuclear pore proteins and known transport factors in *t*-test difference increments (Figure 1A [DOI](#), Figure S1, Table S2). Additionally, we reanalysed previously published data of C-terminally tagged MEX67 and NUP158 in the same way (Moreira et al., 2023, Figure 1A [DOI](#), Figure S1, Table S2). Neither of the bait proteins caused biotinylation of the entire pore complex. Instead, we observed highly specific labelling patterns for each bait. While the central NUP96 biotinylated most nuclear pore proteins with the marked exception of the nuclear basket, NUP158 and NUP76 caused biotinylation of specific sub-complexes of the pore and MEX67 labelled NUPs lining the channel of the pore (Moreira et al., 2023). The data show that the labelling radius of TurboID, under these conditions, is below the size of the nuclear pore and thus well below the resolution of light microscopy.

Next, we imaged cells expressing nuclear pore proteins fused to the TurboID-HA tandem tag using both cy3-streptavidin and anti-HA in combination with a secondary antibody coupled to the Alexa488 fluorophore (Figure 1B [DOI](#)). For *T. brucei* NUP132-TurboID-HA, no obvious differences in resolution were observed between antibody-based and streptavidin-based detection (Figure 1C [DOI](#)). Likewise, for human NUP88, expressed as N-terminal TurboID-4HA fusion protein in HeLa cells, grown in DMEM medium (relying on the biotin of the serum supplement with a low nanomolar concentration), the signal of anti-HA and of streptavidin appeared highly similar (Figure 1D [DOI](#)). Wild type cells of both Trypanosomes and human showed only a very low streptavidin signal, indicating that the signal from naturally biotinylated proteins is neglectable (Figure S8 in supplementary material). The data demonstrate that imaging of Turbo-ID fusion proteins via streptavidin is a reliable alternative, with a resolution that is indistinguishable from antibody-based immunofluorescence in standard light microscopy, which is in agreement with published data on experimentally determined BioID labelling radii (Branon et al., 2018 [DOI](#); Kim et al., 2016 [DOI](#)). The one exception are very motile proteins that produce a “biotinylation trail” distinct to the steady state localisation; these exceptions, and how they can be exploited to understand protein interactions, are discussed in chapter 4 below.

2. TurboID-fusion proteins can be imaged in phase separated/protein-dense regions

We had previously expressed one of the *T. brucei* nuclear mRNA transporters, *Tb*MEX67, fused to TurboID-HA and observed that anti-HA failed to stain MEX67 at the nuclear pores, indicating a possible failure of the antibody to either penetrate phase separated areas or to bind the epitope in this environment (details in introduction, (Moreira et al., 2023), Figure 2A [DOI](#)).

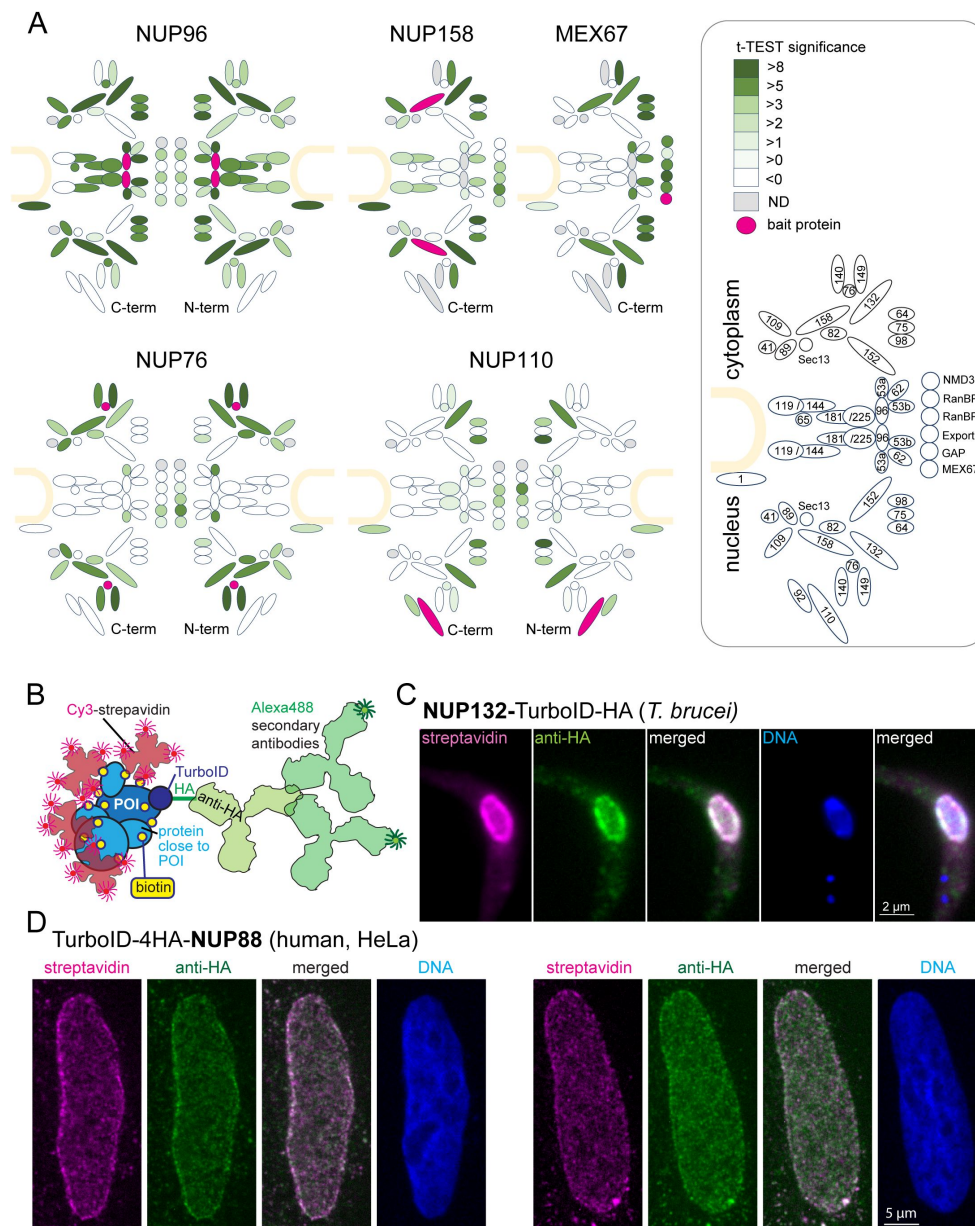


Figure 1

The TurboID biotinylation labelling radius is sufficiently small to allow streptavidin-based imaging of target proteins by light microscopy

A) Schematic representation of the trypanosome NPC (Obado et al., 2016) including selected transport factors and their enrichment observed in TurboID experiments followed by streptavidin affinity capture and LC MS/MS analysis. Proteins quantified are filled in shades of green representative of the corresponding *t*-test difference increments. The respective bait protein is drawn in pink and undetected proteins are in gray. NUP96, NUP158 and NUP76 experiments were analysed with TurboID-HA fusions at both respective termini (as indicated). The nuclear envelope is drawn in sand and nucleoporins and transport factors are numbered in the legend (right) according to (Obado et al., 2016) (NMD3, Tb927.7.970; RanBP1, Tb927.11.3380; RanBPL, Tb927.10.8650; exportin 1, Tb927.11.14340; GAP, Tb927.10.7680).

B) Schematics illustrating the imaging concept for a protein of interest (POI) by either Cy3- streptavidin or anti-HA coupled with secondary Alexa488-labelled antibodies.

(C-D) Nuclear pore proteins of *Trypanosoma brucei* (C) and HeLa cells (D) were expressed fused to the streptavidin-HA tandem tag and detected with streptavidin-cy3 (pink) and anti- HA (green). Representative single plane images of a Z-stack is shown, as raw data.

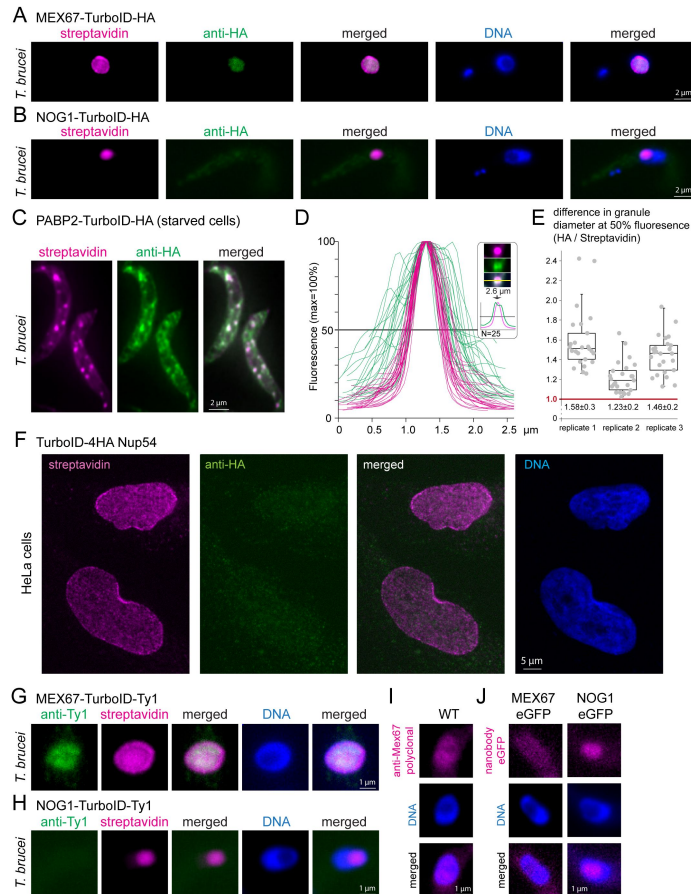


Figure 2

Streptavidin can detect targets within phase-separated regions, while most antibodies fail

(A-B) *T. brucei* MEX67 (A) and *T. brucei* nucleolar protein NOG1 (B) were expressed fused to a C-terminal TurboID-HA tandem tag and cells probed with streptavidin-cy3 (pink) and by anti- HA immunofluorescence (green). Representative single plane images of an unprocessed Z- stack series are shown.

(C-E) Cells expressing the stress granule marker protein PABP2 fused to TurboID-HA were starved (2 hours PBS) and starvation stress granules detected by streptavidin (pink, Cy3) and anti-HA (green, Alexa488). The starvation experiment was performed in biological triplicates. One representative image of starved cells is shown as Z-stack projection (72 slices a 140 nm, sum slices) in C. For each replicate, intensity profiles across one of the larger granules of the cell were measured for 25 cells in both fluorescence channels. The profiles for replicate 1 are shown in D. For each granule, the granule diameter was calculated from the profiles at 50% fluorescence and the difference in diameter between the HA- and streptavidin stain is presented in E for each replicate, as quotient of granule diameters. Note that despite differences between the three replicates, likely arisen from starvation conditions being not 100% reproducible, the HA stain consistently delivered a larger granule diameter than the streptavidin stain, consistent with preferentially peripheral staining of the granule by anti- HA. For replicate 2, the fluorophores were switched, with essentially the same result (Figure S2 in Supplementary Materials).

(F) Human NUP54 fused to TurboID-4HA was expressed in HeLa cells and cells were probed with both anti-HA (green, Alexa488) and streptavidin (Cy3, shown in pink). Streptavidin, but not anti-HA detects NUP54 at the nuclear pores. A single plane image of a Z-stack is shown as raw data.

(G and H) *T. brucei* MEX67 (G) and NOG1 (H) were expressed as TurboID-Ty1 fusion proteins and detected with streptavidin (Cy3, shown in pink) and anti-Ty1 (BB2, green). Representative single plane images of unprocessed Z-stack images are shown.

(I) Trypanosome wild type (WT) cells were probed for MEX67 with polyclonal antiserum (kind gift of Mark Carrington, University of Cambridge; secondary antibody Alexa 488, shown in pink). One representative single plane image of an unprocessed Z-stack image is shown. **(J)** *T. brucei* MEX67 (left) and NOG1 (right) were expressed as eGFP fusion proteins and detected with Cy5 labelled eGFP nanobodies. Representative single plane images of unprocessed Z-stack images are shown.

To test this hypothesis, we focussed on two additional *bona fine* phase-separated regions: the nucleolus and starvation stress granules. We expressed a C-terminal TurboID-HA fusion of the trypanosome nucleolar GTP-binding protein NOG1 (Tb927.11.3120) (Billington et al., 2023 [\[link\]](#)). The resulting streptavidin signal was exclusively at the nucleolus, which is tractable as distinct compartment within the nucleus lacking DAPI fluorescence (**Figure 2B** [\[link\]](#)). In contrast, the HA-signal showed no obvious accumulation in the nucleolus and was extremely weak with a patchy distribution throughout the entire cell body, indicating an unspecific background stain (**Figure 2B** [\[link\]](#)). Thus, just like MEX67, the nucleolar protein NOG1 is detectable by streptavidin, but not by anti-HA immunofluorescence. Next, we starved trypanosome cells that were expressing the established starvation stress granule marker PABP2 (Fritz et al., 2015 [\[link\]](#); Kramer et al., 2013 [\[link\]](#)) fused to TurboID-HA and probed the cells with the streptavidin anti-HA combination. Both anti-HA and streptavidin readily detected starvation stress granules (**Figure 2C** [\[link\]](#)). However, the granules detected by anti-HA were in general less distinct and appeared larger than those detected by streptavidin. Indeed, intensity profiles drawn through Z-stack projections of stress granules were 1.4 ± 0.2 -fold wider for anti-HA detection in comparison to streptavidin detection and the corresponding HA profiles frequently exhibited a double-peak (**Figure 2D and E** [\[link\]](#)). The data are consistent with anti-HA preferentially staining the periphery of the granules, while streptavidin diffuses into and binds within these dense, phase separated particles. A switch of the two fluorophores (Cy3 and Alexa488) gave identical results, controlling for microscopy artefacts as a reason for the differences (Figure S2). Direct eYFP- fluorescence intensity profiles of granules from cells expressing PABP2-eYFP resembled those of the streptavidin stain, ruling out a preferential granule-peripheral localisation of PABP2 as an explanation (Figure S3). Thus, anti-HA fails to label three distinct phase-separated structures of the trypanosome cell, the nuclear pore channel, the nucleolus and the inside of starvation stress granules, while streptavidin faithfully decorates all these structures.

The inability of anti-HA to detect target epitopes within phase-separated areas is not restricted to trypanosomes: When expressing TurboID-4HA fusions of the human FG-repeat nucleoporin NUP54 in HeLa cells, streptavidin detected the proteins at the nuclear pores, while no specific anti-HA signal was observed (**Figure 2F** [\[link\]](#)).

Furthermore, the inability to label phase-separated areas is not restricted to anti-HA but applies to other antibodies. We expressed *T. brucei* MEX67 and NOG1 fused to TurboID-Ty1 and failed to label the nuclear pores and the nucleolus with anti-Ty1 (BB2), respectively, while the streptavidin signal displayed the correct localisations (**Figure 2G and H** [\[link\]](#)). Moreover, MEX67 was undetectable at the nuclear pores by polyclonal antiserum (**Figure 2I** [\[link\]](#); (Pozzi et al., 2023 [\[link\]](#))) and an eGFP-fusion of MEX67 could not be detected by fluorophore labelled eGFP nanobodies at the pores either (**Figure 2J** [\[link\]](#)). The absence of a nanobody signal rules out that its simply the size of IgGs that prevents the staining of Mex67 at the nuclear pores, as nanobodies are smaller than (tetrameric) streptavidin. In contrast, an eGFP fusion of NOG1 was detectable by eGFP nanobodies at the nucleolus, albeit weakly (**Figure 2J** [\[link\]](#)). In conclusion, most but not all antibodies/nanobodies were ineffective in phase separated areas.

3. Boosting the signal under low-antigen conditions (expansion microscopy, CLEM)

A potential advantage for imaging TurboID fusion proteins with streptavidin is an increased signal: TurboID is expected to cause multiple lysine-biotinylation of both the bait protein and proteins in close proximity, while antibodies only decorate several (polyclonals) or one specific (monoclonals) epitope of a target protein. Indeed, for many TurboID-HA fusion proteins that we analysed by standard light microscopy, the streptavidin signal appeared significantly stronger and devoid of background stain when comparing to anti-HA. To quantify this observation, we probed cells expressing NUP158-TurboID-HA with combinations of anti-HA/Alexa488 secondary antibody and Streptavidin-Cy3 (**Figure 3A** [\[link\]](#)) or with Streptavidin-Alexa488 and anti-HA/Alexa594

secondary antibody (**Figure 3B** [↗](#)). Under both labelling conditions, the streptavidin stain resulted in significantly less non-specific cytoplasmic background signal than anti-HA (**Figure 3A-B** [↗](#)). Moreover, the maximum Alexa488 signal of Z-stack projections was 2.9-fold higher for streptavidin than for anti-HA (N=60, unpaired two-tailed t -test=9.6E-55) (**Figure 3C** [↗](#)). An increase in fluorescence signal is desirable for the detection of low abundance proteins, but also for imaging applications that necessitate a reduction in antigen density, such as expansion microscopy or CLEM.

In expansion microscopy, biomolecules are covalently cross-linked to a hydrogel, that is subsequently swollen by osmosis, causing a physical magnification of the sample (Chen et al., 2015 [↗](#)). Even a relatively small expansion factor of 3 causes a 27-fold increase in sample volume and thus a 27-fold decrease in antigen-density. The two main expansion microscopy methods are Protein-retention Expansion Microscopy (proExM) (Tillberg et al., 2016 [↗](#)) and Ultra-structural Expansion Microscopy (U-ExM) (Gambaretto et al., 2019 [↗](#)). The main difference is that in proExM, antibody labelling is applied prior to the expansion process, while in U-ExM it is done afterwards. U-ExM thus generally results in higher resolution, as the linkage-error (difference between position of the fluorophore of the secondary antibody and the position of the target protein) is not expanded. Next to the decrease in antigen-density as a consequence of the expansion, the signal is further compromised by the harsh conditions associated with the method, as for example an incubation step at 95°C in alkaline SDS-containing buffer for 90 min for U-ExM and an over-night proteinase K digestion for ProExM. To assess the potential of streptavidin imaging, we employed both proExM and U-ExM on cell lines expressing TurboID-HA fusions of two trypanosome nuclear-pore localised proteins (NUP76 and MEX67) and compared the signals from streptavidin and anti-HA (**Figure 3D** [↗](#)). For both proteins, streptavidin caused a strong and uniform stain of all nuclear pores with both expansion microscopy methods. In contrast, anti-HA failed to detect nuclear pores either completely (MEX67) or partially (NUP76) in U-ExM. In proExM, pores could be stained evenly with anti-HA, but the signal was much lower in comparison to streptavidin, in particular for NUP76. To our knowledge, this is the first time that streptavidin imaging of a biotin ligase-fused protein was used in combination with expansion microscopy. In our hands, streptavidin clearly outperformed antibodies in both expansion protocols and with both target proteins that we tested, giving a significantly stronger signal with less background.

CLEM causes an even larger reduction in accessible antigen than expansion microscopy. Thin slices of a resin-embedded sample are first used for immunofluorescence and imaged by light microscopy, and, in a second step, imaged by electron microscopy. The resin largely prevents antibody penetration and probing is therefore restricted to the small fraction of antigens present at the surface of the resin slice. Trypanosome cells expressing NUP96- TurboID-HA or NUP158- TurboID-HA were embedded in LR-white and slices were probed with streptavidin and anti-HA. For both proteins, the streptavidin signal detected numerous pores for each nucleus, while anti-HA detected between 0 and 2 pores and, in addition, caused considerably higher background signals (**Figure 3E** [↗](#) and Figure S4 in Supplementary materials). Streptavidin is thus more suitable for CLEM, and an example, targeting NUP158- TurboID-HA with both steps, is shown in **Figure 3F** [↗](#).

In summary, streptavidin detection causes a massive increase in signal that is in particularly beneficial under low-antigen density conditions, and thus offers significant implications in expansion microscopy and CLEM.

4. Visualization of protein interactions

We found that in most cases, streptavidin labelling faithfully reflects the steady state localisation of a bait protein, e.g., the localisation resembles those observed with immunofluorescence or direct fluorescence imaging of GFP-fusion proteins. For certain bait proteins, this is not the case, for example, if the bait protein or its interactors have a dynamic localisation to distinct compartments, or if interactions are highly transient. It is thus essential to control streptavidin-based *de novo* localisation data by either antibody labelling (if possible) or by direct fluorescence

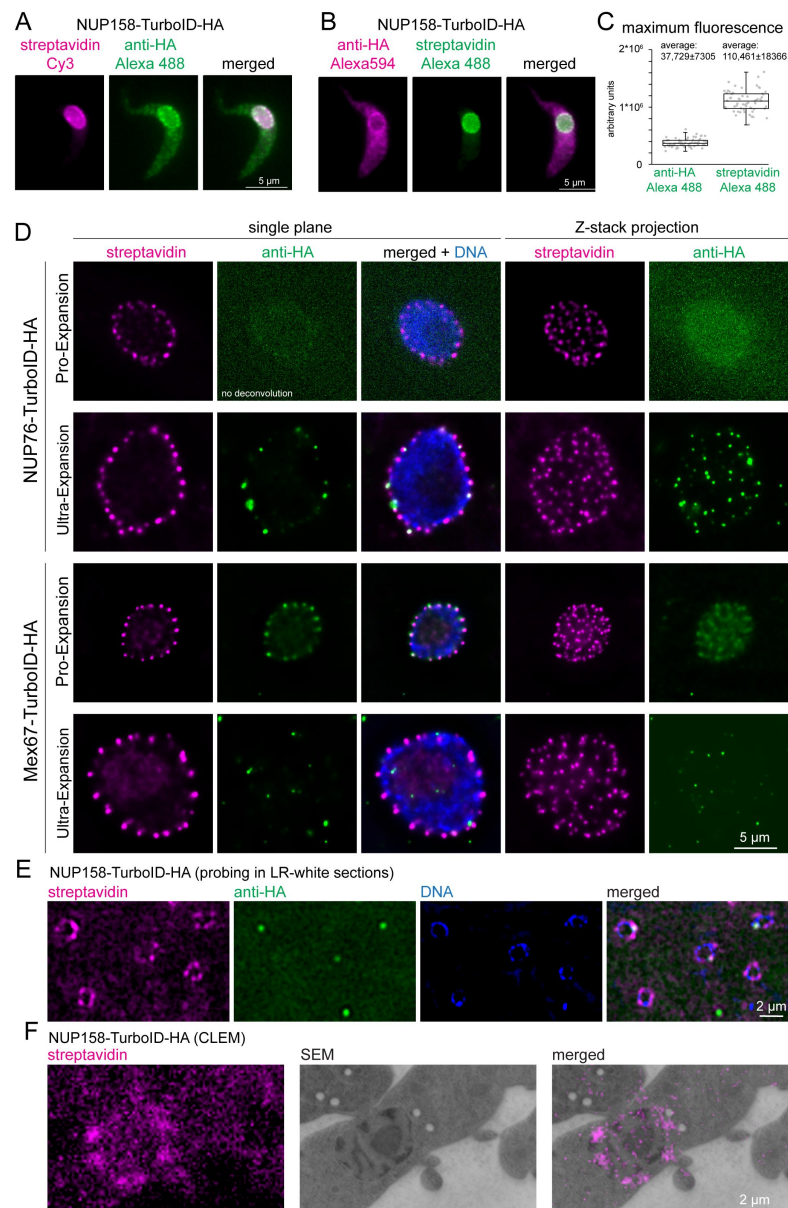


Figure 3

Streptavidin imaging yields higher signal intensities than immunofluorescence

(A-C) Enhanced signal in standard light microscopy

Trypanosome cells expressing NUP158-TurboID-HA were labelled with combinations of either streptavidin-Cy3 and anti-HA/Alexa488 secondary (A) or with streptavidin-Alexa488 and anti-HA/Alexa 594 secondary (B). Z-stack images were recorded (48 slices a 140 nm). Representative, unprocessed single plane images are shown (A and B). The maximum Alexa488 fluorescence was quantified from Z-stack projections (sum slices) from 60 cells probed with anti-HA or streptavidin; the data are presented as a dot blot (waist is median; box is IQR; whiskers are ± 1.5 IQR) (C)

(D) Improved signal in expansion microscopy

Trypanosome cells expressing NUP76-TurboID-HA or MEX67-TurboID-HA were imaged using Pro-expansion or Ultra-expansion microscopy. Single plane and Z-stack projections (sum slices) of the streptavidin and anti-HA signal are shown for one representative nucleus. All images were deconvolved in proExM, except for NUP76_{anti-HA}.

(E-F) Improved signal in CLEM

Trypanosome cells expressing NUP158-TurboID-HA were embedded in LR-White resin. Slices were probed with streptavidin and anti-HA and imaged by light microscopy (E) followed by electron microscopy (CLEM) (F).

of fusion-proteins for each new bait protein. However, any additional signal from biotinylated partner proteins at a different location can also be leveraged to gain relevant information on the function and dynamics of the bait protein. Three respective examples are detailed below:

The first is PABP2, one of two trypanosome poly(A) binding proteins, that has established, unambiguous cytoplasmic localisation, when visualised with either anti-HA or when expressed fused to a fluorescent protein (**Figure 4A-B** [↗](#), [Billington et al., 2023](#) [↗](#); [Kramer et al., 2013](#) [↗](#)). However, when PABP2-TurboID-HA expressing cells are probed with streptavidin, a signal at the posterior pole of the cell is observed, in addition to the expected cytoplasmic signal (**Figure 4A** [↗](#)). Localisation to the posterior pole is an exclusive feature of the trypanosome mRNA decapping enzyme ALPH1 and its four interaction partners: a small proportion of this mRNA decapping complex localises to the posterior pole, in a dynamic manner, while the majority remains cytoplasmic ([Kramer, 2017](#) [↗](#); [Kramer et al., 2008](#) [↗](#); [Kramer et al., 2023](#) [↗](#)) (**Figure 4B** [↗](#)). The streptavidin signal at the posterior pole of the PABP2-TurboID-HA cells thus likely reflects a historic cytoplasmic interaction between PABP2 and the mRNA decapping complex. The existence of this interaction is supported by the identification of all five posterior-pole localised members of the decapping complex by mass spectrometry data from a PABP2 BioID experiment ([Moreira et al., 2023](#)). Notably, imaging with streptavidin adds information about protein dynamic interactions, that mass spectrometry data alone cannot uncover. In this case, it shows that the fraction of mRNA decapping proteins at the posterior pole is dynamically exchanged with the fraction in the cytoplasm.

The second example is MLP2 (also called NUP92), a divergent nucleoporin of Kinetoplastida involved in chromosome distribution during mitosis ([Holden et al., 2014](#) [↗](#); [Morelle et al., 2015](#) [↗](#)). MLP2 localisation is cell cycle dependent: while exhibiting a nuclear localisation during interphase, it migrates to the spindle and spindle pole during mitosis ([Holden et al., 2014](#) [↗](#); [Morelle et al., 2015](#) [↗](#)) (**Figure S5**). We expressed MLP2 fused to an N-terminal TurboID-HA tag and examined the streptavidin and HA signal in interphase and mitotic cells (**Figure 4C** [↗](#)). Note that in trypanosomes the kinetoplast (mitochondrial DNA, visible as small dot in DAPI stain) divides prior to the nucleus and numbers and positions of kinetoplasts and nuclei are established markers for cell cycle stages ([Sherwin and Gull, 1989](#) [↗](#)). We observed that the streptavidin signal “lagged behind” the anti-HA signal at all cell cycle stages. This is most obvious in anaphase, when anti-HA exclusively stains the spindle pole body, while streptavidin decorates both, the nuclei and the spindle pole body. Thus, for proteins with cell-cycle dependent localisation, streptavidin-labelling provides information on both, the previous localisation(s) of the bait, in addition to its current position.

The third example are nuclear pore proteins. For all NUPs that we tested, we observed the expected streptavidin signal at the nuclear pores (see below). However, for some NUPs, for example NUP65, we observed additional cytoplasmic streptavidin stain that is absent for others, as for example NUP75 (**Figure 4 D-E** [↗](#)). There were also differences in the amount of nucleoplasmic biotinylation between NUPs. For example, NUP75 showed more nuclear labelling than NUP65 (**Figure 4 D-E** [↗](#)). The likeliest explanation is, that some NUPs are contacting and thus proximity-labelling proteins in transit. The streptavidin stain in the nucleoplasm or cytoplasm thus provides information about the functions of the individual NUPs in protein import or export, respectively.

Importantly, tracking of proteins by streptavidin imaging requires orthogonal controls, as the imaging alone does not provide information about the nature of the biotinylated proteins. These can be proximity ligation assay, mass spectrometry or specific tagging visualisation of protein suspects by fluorescent tags. Once these orthogonal controls are established for a specific tracking, streptavidin imaging is an easy and cheap and highly versatile method to monitor protein interactions in a specific setting.

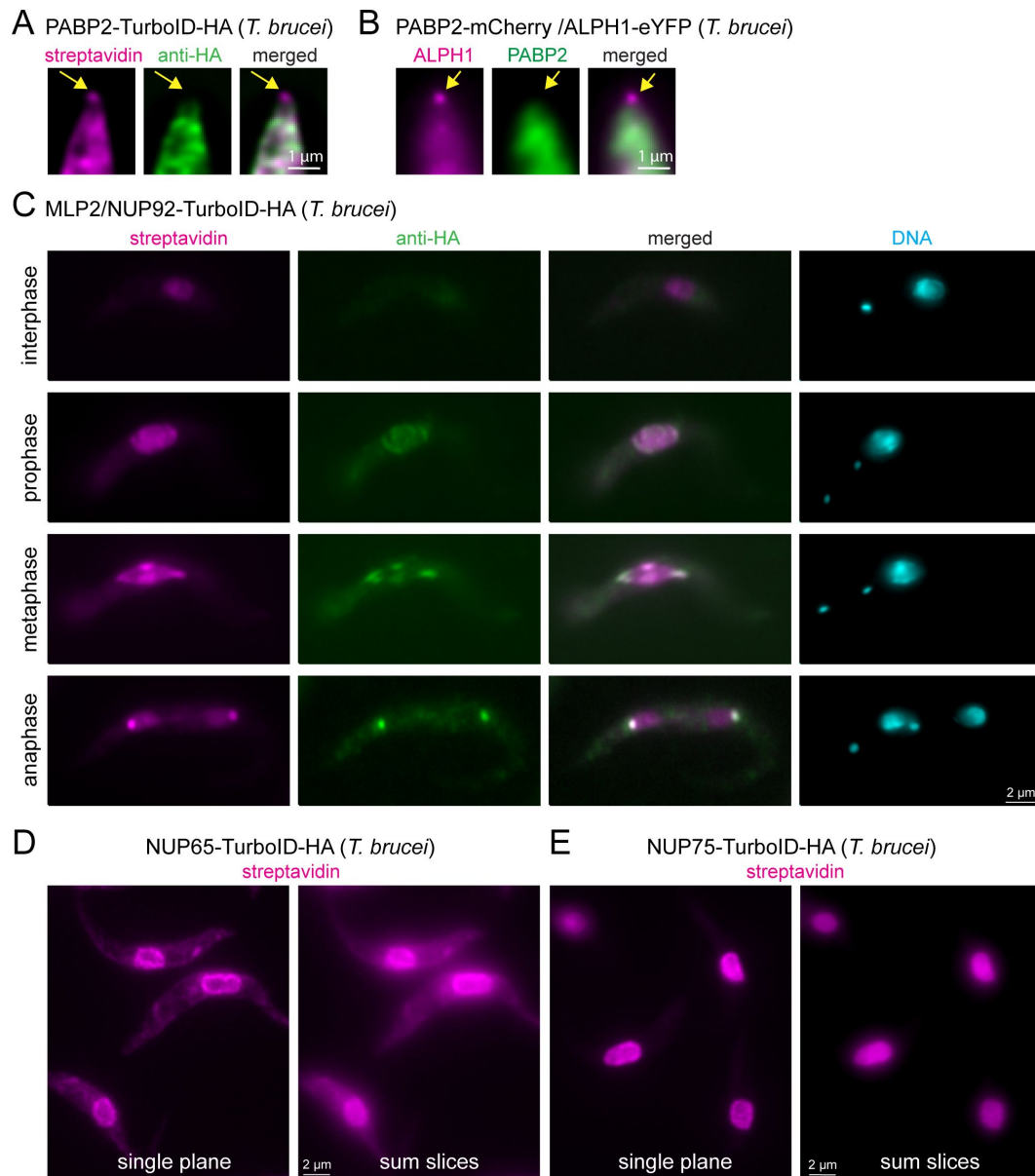


Figure 4

Visualisation of protein interactions with TurboID

(A) Trypanosome cells expressing PABP2-TurboID-HA were probed with streptavidin and anti-HA. A representative image of the posterior part of a cell is shown (single plane of a Z-stack processed by computational clearing). The position of posterior pole granule is indicated by arrows.

(B) Trypanosome cells were transformed to co-express PABP2-mChFP and ALPH1-eYFP. A representative image of the posterior part of a cell is shown (single plane of a Z-stack processed by deconvolution). The position of posterior pole granule is indicated by arrows. **(C)** Trypanosome cells expressing TurboID-HA-MLP2 were probed with streptavidin and anti-HA. One representative image of an interphase, prophase, metaphase and an anaphase cell are shown. All images are unprocessed single plane images, with the exception of the DAPI image that is a Z-stack projection (max intensity of 48 slices a 140 nm).

(D and E) Trypanosome cells expressing NUP65 (D) or NUP75 (E) fused to a C-terminal TurboID-HA tag were probed with streptavidin. Single plane and Z-stack projection (sum slices of 48 slices a 140 m) of unprocessed images are shown.

5. A phase-separation map of the *T. brucei* nuclear pore

We reasoned that our findings that all phase-separated regions of trypanosomes that we tested are inaccessible to anti-HA can be exploited as a tool for de novo identification of phase-separated regions. As a proof of principle, we employed streptavidin-imaging to produce a map of antibody-inaccessible regions of the entire trypanosome nuclear pore complex. We endogenously expressed all known trypanosome nuclear pore proteins fused to TurboID-HA from one allele. For each protein, two cell lines were constructed, one with TurboID-HA fused to the C-terminus, and one with the tag at the N-terminus. We excluded Mlp2/Nup92, as we found this protein localizing to the nucleus rather than the nuclear pores in most interphase cells (Figure S5 in Supplementary material and **Figure 4C**), in agreement with (Morelle et al., 2015). For two C-terminal fusion candidate proteins (Sec13, NUP140) and one N-terminal (NUP82), we repeatedly failed to obtain cell lines. The remaining 45 cell lines were controlled by western blot probed with anti-HA to confirm the correct size of the fusion protein (Figure S6). Next, cells from all cell lines were probed with anti-HA and fluorescent streptavidin. All cell lines showed streptavidin signal predominantly at the nuclear pores, proving the correct localisation of the fusion proteins (**Figure 5A**). For N-terminally tagged NUP64 we also observed between one and two streptavidin dots in the nucleus, consistent with the localisation of eYFP-NUP64 in life cells (Billington et al., 2023); a likely mislocalisation.

When evaluating the HA signal at the nuclear pores, we observed no antibody stain for the NUPs lining the central channel, while most NUPs at the periphery of the channel were accessible to the antibody (**Figure 5B**). Importantly, there is no correlation between protein abundance, as determined by proteomics (Tinti and Ferguson, 2022), and immunofluorescence signal, ruling out low antigen abundance as the reason for its absence (Figure S7). All the FG-NUPs of the inner channel (which are known to phase-separate) were among the non-accessible proteins, supporting the idea that the crucial determinant preventing anti-HA detection is mainly or entirely phase separation. Likewise, the mRNA-interacting NUP76 complex consisting of NUP76 and the FG NUPs NUP140 and NUP149 appear non-accessible to anti-HA, except when NUP149 is tagged at the N-terminus. The NUP76 interacting protein NUP158 was inaccessible to antibodies when tagged at the N-terminus, but not at the C-terminus: this is consistent with NUP158 being partially phase-separated as it contains FG-repeats exclusively at the N-terminal half. Among the antibody inaccessible proteins were also some non-FG NUPs. These include proteins of the inner ring, NUP181, NUP144, C-terminally tagged NUP225 and N-terminally tagged NUP65, as well as N-terminally tagged proteins of the outer ring, NUP89 and NUP41. Whether the respective termini of these proteins extend into phase separated areas, or whether antibody access is prevented by steric or other hindrances remains unknown.

In conclusion, we here provide an antibody access map of the entire *T. brucei* nuclear pore that includes all bone-fine FG-NUPs in addition to some non-FG NUPs.

Discussion

We here show that the use of the TurboID-HA tandem tag offers an “all in one” solution to antibody-accessibility problems, without compromising resolution or the strength of the signal. Quite the contrary, streptavidin imaging of TurboID-HA tagged proteins facilitates a massive boost in signal, extending the benefits of the method to protein targets with low antigen abundance. Furthermore, the TurboID-HA tag is suitable to probe protein interactions and provide some evidence for phase-separation.

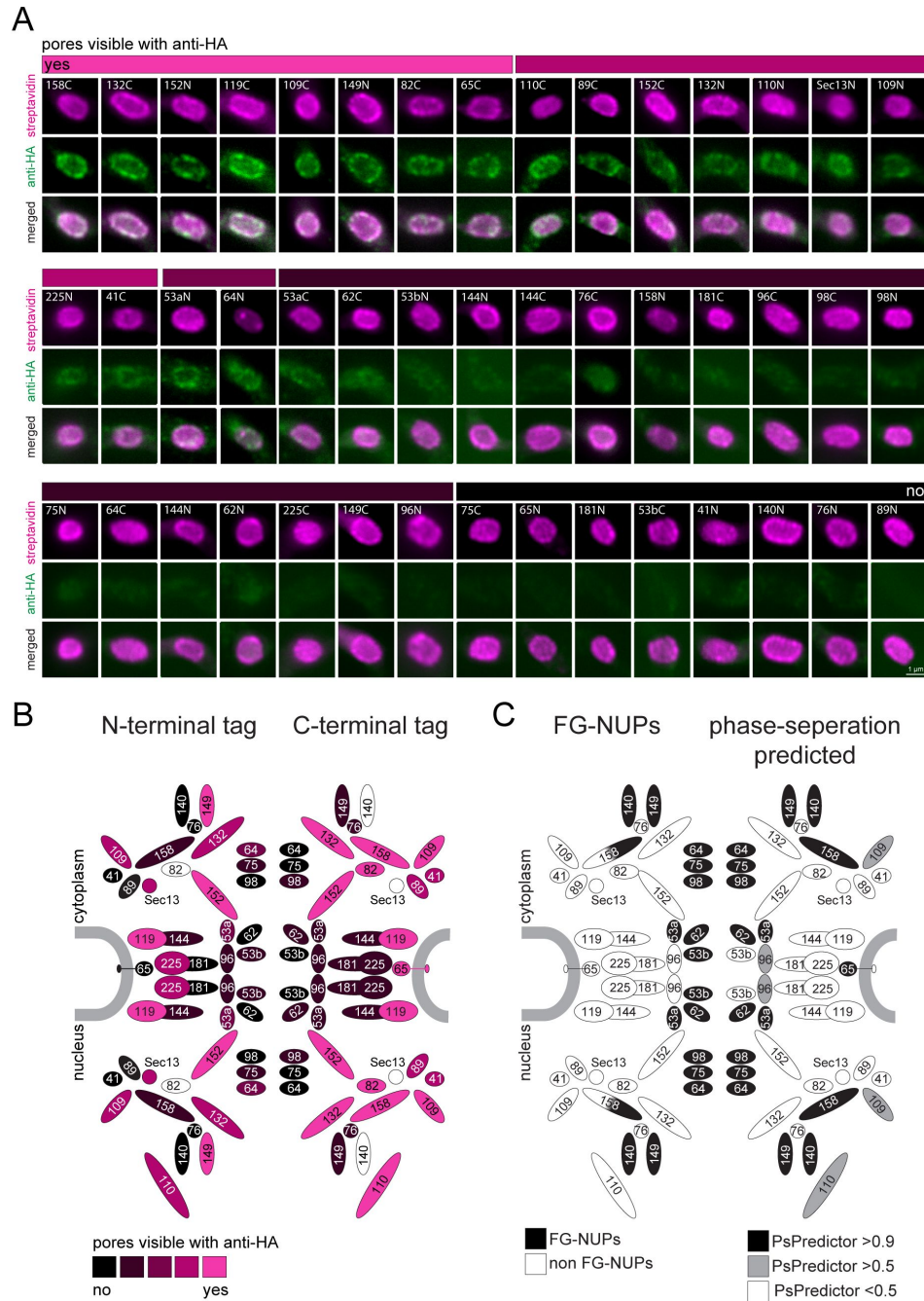


Figure 5

A refined map of the *T. brucei* nuclear pore complex

Each known *T. brucei* nuclear pore protein was expressed fused to TurboID-HA, each both at the N- and C-terminus. Cells were labelled with anti-HA and with streptavidin.

(A) For each NUP, a representative image of the nucleus is shown as unprocessed, single plane of a Z-stack. We evaluated the extent of anti-HA stain in colour increments (shown as a bar above images).

(B) The HA-signal at the nuclear pores was mapped onto a schematic representation of the trypanosome NPC (modified from (Obado et al., 2016)) using the same colour increments. **(C)** Scheme of the *T. brucei* nuclear pore, with known FG NUPs (Obado et al., 2016) shown in black on the left side and prediction of phase separation (Chu et al., 2022) mapped on the right.

Like many methods that are frequently used in cell- and molecular biology, streptavidin imaging is based on the expression of a genetically engineered fusion protein: it is essential to validate both, function and localisation of the TurboID-HA tagged protein by orthogonal methods. If the fusion protein is non-functional or mis-localised, tagging at the other end may help, but if not, this protein cannot be imaged by streptavidin imaging. Likewise, target organisms not amenable to genetic manipulation, or those with restricted genetic tools, are not or less suitable for this method.

One protocol suits all - streptavidin readily traces biotin proximity labels

Many proteins are not accessible to antibodies in standard immunofluorescence protocols, in particular proteins localized in phase-separated regions. MEX67 could not be detected at the trypanosome nuclear pore by anti-HA, anti-Ty1, anti-Protein A, polyclonal MEX67 antibodies and even nanobodies to mNeonGreen (this work, [\(Dostalova et al., 2013\)](#); [Moreira et al., 2023](#); [Pozzi et al., 2023](#)); likewise, the human Mex67 homologue TAP/Nxf1 evaded detection by antiserum [\(Bear et al., 1999\)](#). HA-fused trypanosome FG-NUPs and the human FG-NUP NUP54 could not be decorated with anti-HA. Trypanosome starvation stress granules could only be labelled at the periphery and nucleolar proteins evaded antibody detection in trypanosomes (**Figure 2**) and in human cells [\(Misteli, 2008\)](#); [Musinova et al., 2011](#); [Sheval et al., 2005](#); [Zatsepina et al., 1997](#)). There are several approaches available to trouble-shoot antigen-accessibility problems [\(Piña et al., 2022\)](#). Pre-extraction steps prior to fixation employing Triton X100, high salt or nucleases proved successful for the detection of mammalian Mex67 homologue TAP/NXF1 [\(Ben-Yishay et al., 2019\)](#); [Katahira et al., 1999](#)). Antibody penetration to nucleolar proteins was improved by including protease treatment [\(Svistunova et al., 2011\)](#). Further, fixation by cold methanol instead of paraformaldehyde was reported to increase antibody access to proteins [\(Neuhaus et al., 1998\)](#). However, the search for individual solutions for each problematic antibody/antigen pair is cumbersome and an improvement of antibody access typically comes at the expense of signal loss and/or disruption of cellular morphology [\(Piña et al., 2022\)](#). With streptavidin labelling we present a method that works for every protein (we tested ~30), without compromising cellular morphology and, most importantly, provides a strong signal as significant additional benefit.

In standard light microscopy, streptavidin imaging offers a resolution comparable to labelling with antibodies. The labelling radius of the biotin ligase BirA was estimated to be approximately 10 nm by using the NUP107-160 Y-complex of the mammalian nuclear pore as a molecular ruler [\(Kim et al., 2014\)](#), enabling for example the determination of substructures of P-bodies and stress granules [\(Youn et al., 2018\)](#). It is thus within the range of an average-sized protein and well below the resolution limit of light microscopy. Consistently, cells expressing nuclear membrane associated proteins fused to BirA (or variants as the related BioID2) can be probed by fluorescent streptavidin with resolutions that are undistinguishable from probing with the respective IgG antibodies [\(Kim et al., 2016\)](#); [May et al., 2020](#). For the biotin ligase variant TurboID, which has a massively decreased labelling time, the biotin labelling radius was measured to be in a similar [\(Branon et al., 2018\)](#); [Kim et al., 2016](#) or slightly larger range [\(May et al., 2020\)](#). Importantly, the addition of 50 μ M external biotin massively increased the labelling radius, while in the absence of excessive biotin, the labelling radius remained comparable to the labelling radius of BirA [\(May et al., 2020\)](#). In this work, we avoided the addition of additional biotin to the trypanosome medium (the medium contains 0.8 μ M biotin) and serum biotin (approximately 2 nM in mammalian serum [\(Luong and Vashist, 2020\)](#)) was the sole source in HeLa cell medium. Our mass spectrometry data of streptavidin-affinity purified proteins with five nuclear pore localised TurboID-fusion proteins demonstrates that the labelling radius is well below the size of the nuclear pore complex under these conditions, and thus in agreement with published data. Even in expansion microscopy, with 3.6 to 4.2 -fold expansion factors, we did not notice any decrease in resolution; in fact, we could resolve NUPs that reside at different positions within the pore (e.g.

inner ring vs basket) by labelling one with streptavidin and the other with an antibody and thus obtained sub-NUP imaging resolution (our own unpublished data). Even though, theoretically, BioID *per se* causes a decrease in resolution as biotinylation extends to proteins in close proximity to the bait, this may well be compensated by the lower linkage error when imaging directly via the small, compact-shaped streptavidin-tetramer rather than by a tandem assembly of two large IgG antibodies (**Figure 1B** [↗](#)).

Streptavidin tracing of biotin proximity labels delivers high signal intensities

One major benefit of streptavidin imaging was the increase in signal compared to immunofluorescence, caused by multiple biotinylation sites of the bait and adjacent proteins serving as an enhancer. An increased signal is beneficial for almost all protein-imaging applications, but in particular for low abundance targets. Most strikingly, streptavidin imaging facilitates imaging targets with low densities as a consequence of the imaging method, as in expansion microscopy and CLEM and is applicable to all target proteins that tolerate tag fusion to either terminus. Fluorescent streptavidin has been previously used in expansion microscopy to detect biotin residues in target proteins produced by click chemistry (Sun et al., 2021 [↗](#)). However, to the best of our knowledge, this is the first report that employs fluorescent streptavidin as a signal enhancer in expansion microscopy and CLEM, by combining it with multiple biotinylation sites added by a biotin ligase. Importantly, for both CLEM and expansion, streptavidin imaging is the only alternative approach to immunofluorescence, as denaturing conditions associated with these methods rule out direct imaging of fluorescent tags.

A combination tag as probe for phase separated areas

A further important application of the TurboID-HA tandem tag is the *de novo* prediction of phase separated areas in cells. The importance of compartmentalization by liquid-liquid phase separation for almost all cellular functions, including DNA and mRNA metabolism (Dai and Yang, 2023 [↗](#)), cell division (Ong and Torres, 2020 [↗](#)) and development (So et al., 2021 [↗](#)) becomes increasingly evident. Aberrant phase separation can cause disease, as for example neurodegenerative disorders (Chakraborty and Zweckstetter, 2023 [↗](#); Tsoi et al., 2023 [↗](#)), underlining the relevance of novel tools for identification and characterization of phase separated regions. For individual proteins, phase separation can be predicted in silico using machine learning, with limited, but increasing accuracy (Venko and Žerovnik, 2023 [↗](#)). For a defined mixture of molecules, phase separation can be tested in vitro (Alberti et al., 2018 [↗](#); Zhang and Shen, 2023 [↗](#)). However, methods to test phase separation in the cellular context are scarce. The correlation between antibody-inaccessibility and phase separation, that we observed in this study, offers the opportunity to use the TurboID-HA tag to probe for phase separation *de novo*. Notably, the streptavidin stain concomitantly provides a convenient intrinsic control to confirm correct localization of the fusion protein. Moreover, proteins with complex localization patterns can be simultaneously detected in both, phase-separated and non-phase separated regions, as we demonstrated for tracing MEX67 to both, the nucleoplasm and nuclear pores (**Figure 1A and G** [↗](#)). In trypanosomes, all *bonafide* phase-separated regions were inaccessible to anti-HA, namely the nucleolus, starvation stress granules and the FG NUP environment. The data strongly suggest that the absence of an HA immunofluorescence signal correlates with phase-separation. However, we cannot rule out the possibility, that some proteins may be inaccessible to anti-HA for different reasons, as for example steric hindrance. The latter scenario would equally favor streptavidin over IgG antibodies for target protein binding due to its much smaller dimensions and higher binding affinity. We observed some non-FG Nups to be anti-HA inaccessible, which were mostly NUPs of the inner ring, close to the core of FG NUPs: whether these NUPs localize to phase-separated areas is not known. Importantly, problems to detect phase-separated proteins by antibodies appear conserved across species (Bear et al., 1999 [↗](#); Misteli, 2008 [↗](#); Musinova et al.,

2011 [↗](#); Sheval et al., 2005 [↗](#); Zatssepina et al., 1997 [↗](#)). We show data for the human FG-NUP Nup54 as an example of an anti-HA-inaccessible protein (**Figure 2F** [↗](#)) as proof of principle, that the *de novo* identification of phase-separation is applicable to human cells.

Why are most antibodies and, to some extent even fluorophore conjugated nanobodies, prevented from labelling proteins in phase separated environments, while streptavidin never fails? We cannot distinguish whether the lack in labelling is caused by an access problem of the antibody penetrating the phase separated area or by a reduced binding affinity between antibody and antigen in phase-separated environments, or by a combination of both. Still, the smaller size of streptavidin in comparison to an IgG tandem-antibody pair (**Figure 1b** [↗](#)) likely favors its access into dense regions. Further, IgG antibodies as well as single chain antibodies typically have affinities in the nanomolar range, comparing to femtomolar affinity of the biotin-streptavidin interaction, which is indeed among the strongest non-covalent interactions in nature (Chivers et al., 2011 [↗](#)). Together with the extremely high structural stability of streptavidin towards temperature, pH and denaturing conditions (Laitinen et al., 2006 [↗](#)) this can explain the ability of this probe to potentially outperform antibodies in phase separated environments.

Proximity-labelling reports a history of dynamic protein interactions

Lastly, we propose streptavidin imaging as a novel tool to monitor dynamic interactions between proteins. Since BioID proximity labeling is not restricted to the bait protein, but extends to all interacting partners, the history of interactions is preserved. Hence, changes in protein sub-complexes can be imaged over time. We provide three examples, covering protein complex dynamics in cell cycle regulation and mRNA metabolism as well as nuclear transport. There are many other potential applications. While we relied on an endogenous, constitutive expression of TurboID fusion proteins, a tightly controlled, inducible system, as for example Split-TurboID (Cho et al., 2020 [↗](#)) in combination with light activation (Chen et al., 2022 [↗](#); Shafriz et al., 2023 [↗](#)) would allow pulse chase experiment, delivering quantitative data on the kinetics of processes.

Altogether, streptavidin imaging is a highly versatile tool with multiple applications, ranging from a simple boost in signal that can be leveraged to increase imaging sensitivity in expansion microscopy and CLEM, to the ability to probe for phase-separated areas and to monitor dynamic protein interactions. It is thus an important addition to the available imaging toolbox.

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Reviewer #1 (Public Review):

I feel that the changes to the manuscript have significantly improved it. It's unfortunate that the single biotin/anti-biotin antibodies were not more illuminating but I think the attempts were worthwhile. My only comment is that the rebuttal to the second part of point 3 still does not fully deal with the issue. By marking proximal proteins other than the fusion with biotin,

TurboID significantly increases the detectable signal, but it is formally no longer possible to be certain what the biotin is attached to. None of the controls that the authors suggest will actually give you certainty about what you are detecting while imaging. Mass spectrometry will give you an ensemble measurement of all the biotinylated proteins in the cell without being able to relate that back to what you are observing in a specific cellular region when you are imaging. Colocalization with a tagged protein/specific antibody could suggest that a portion of the signal could be attributable to the TurboID-biotin signal, but it could also be a tight binding partner or part of a larger protein complex. PLA assays would have similar issues- some of the protein could be labeled but it will be impossible to show what portion of the signal is attributable to the protein of interest and how much is attributable to other proximal proteins. I think the key thing here is that in this implementation, TurboID allows you to enhance the labeling of protein structures in cells, such as NUPS, but at the expense of certainty about the specific proteins you are labeling. I personally cannot think of a reasonable control that will allow you to avoid this issue. I feel that this point needs to be clearly made if people are going to use this method for signal enhancement, otherwise people may be misled about what they are actually looking at. The method should be useful, but the limitations need to be clear.

<https://doi.org/10.7554/eLife.95028.2.sa2>

Reviewer #2 (Public Review):

I found the original paper to be of high quality and value. The revisions the authors have made (particularly with respect to the more cautious phraseology concerning the ability to track labelled proteins) are valuable additions. The other responses are well-argued and satisfactory to this reviewer.

<https://doi.org/10.7554/eLife.95028.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

In this work, Odenwald and colleagues show that mutant biotin ligases used to perform proximity-dependent biotin identification (TurboID) can be used to amplify signal in fluorescence microscopy and to label phase-separated compartments that are refractory to many immunofluorescence approaches. Using the parasite Trypanosoma brucei, they show that fluorescent methods such as expansion microscopy and CLEM, which require bright signals for optimal detection, benefit from the elevated signal provided by TurboID fusion proteins when coupled with labeled streptavidin. Moreover, they show that phase-separated compartments, where many antibody epitopes are occluded due to limited diffusion and potential sequestration, are labeled reliably with biotin deposited by a TurboID fusion protein that localizes within the compartment. They show successful labeling of the nucleolus, likely phase-separated portions of the nuclear pore, and stress granules. Lastly, they use a panel of nuclear pore-TurboID fusion proteins to map the regions of the T. brucei nuclear pore that appear to be phase-separated by comparing antibody labeling of the protein, which is susceptible to blocking, to the degree of biotin deposition detected by streptavidin, which is not.

Strengths:

Overall, this study shows that TurboID labelling and fluorescent streptavidin can be used to boost signal compared to conventional immunofluorescence in a manner similar to tyramide amplification, but without having to use antibodies. TurboID could prove to be a viable general strategy for labeling phase-separated structures in cells, and perhaps as a means of identifying these structures, which could also be useful.

Weaknesses:

However, I think that this work would benefit from additional controls to address if the improved detection that is being observed is due to the increased affinity and smaller size of streptavidin/biotin compared to IgGs, or if it has to do with the increased amount of binding epitope (biotin) being deposited compared to the number of available antibody epitopes. I also think that using the biotinylation signal produced by the TurboID fusion to track the location of the fusion protein and/or binding partners in cells comes with significant caveats that are not well addressed here, mostly due to the inability to discern which proteins are contributing to the observed biotin signal.

To dissect the contributions of the TurboID fusion to elevating signal, anti-biotin antibodies could be used to determine if the abundance of the biotin being deposited by the TurboID is what is increasing detection, or if streptavidin is essential for this.

We agree with the reviewer, that it would be very interesting to distinguish whether the increase in signal comes from the multiple biotinylation sites or from streptavidin being a very good binder, or perhaps from both. However, this question is very hard to answer, as antibodies differ massively in their affinity to the antigen which is further dependent on the respective IF-conditions, and are therefore not directly comparable. Even if anti-biotin gives a better signal than anti-HA, this can be either caused by the increase in antigen-number (more biotin than HA-tag) or by the higher binding affinity, or by a combination of both, thus hard to distinguish. Nevertheless, we have tested monoclonal mouse anti-biotin targeting the (non-phase-separated) NUP158. We found the signal from the biotin-antibody to be much weaker than from anti-HA, indicating that, at least this particular biotin antibody, is not a very good binder in IF.

Alternatively, HaloTag or CLIP tagging could be used to see if diffusion of a small molecule tag other than biotin can overcome the labeling issue in phase-separated compartments. There are Halo-biotin substrates available that would allow the conjugation of 1 biotin per fusion protein, which would allow the authors to dissect the relative contributions of the high affinity of streptavidin from the increased amount of biotin that the TurboID introduces.

This is a very good idea, as in this case, the signals are both from streptavidin and are directly comparable. We expressed NUP158 with HaloTag and added PEG-biotin as a Halo ligand. However, PEG-biotin is poorly cell-permeable, and is in general only used on lysates. In trypanosomes, cell permeability is particularly restricted, and even Halo-ligands that are considered highly cell-penetrant give only a weak signal. Even after over-night incubation, we could not get any signal with PEG-biotin. Our control, the TMR-ligand 647, gave a weak nuclear pore staining, confirming the correct expression and function of the HaloTag-NUP158.

The idea of using the biotin signal from the TurboID fusion as a means to track the changing localization of the fusion protein or the location of interacting partners is an attractive idea, but the lack of certainty about what proteins are carrying the biotin signal makes it very difficult to make clear statements. For example, in the case of

TurboID-PABP2, the appearance of a biotin signal at the cell posterior is proposed to be ALPH1, part of the mRNA decapping complex. However, because we are tracking biotin localization and biotin is being deposited on a variety of proteins, it is not formally possible to say that the posterior signal is ALPH1 or any other part of the decapping complex. For example, the posterior labeling could represent a localization of PABP2 that is not seen without the additional signal intensity provided by the TurboID fusion. There are also many cytoskeletal components present at the cell posterior that could be being biotinylated, not just the decapping complex. Similar arguments can be made for the localization data pertaining to MLP2 and NUP65/75. I would argue that the TurboID labeling allows you to enhance signal on structures, such as the NUPs, and effectively label compartments, but you lack the capacity to know precisely which proteins are being labeled.

We fully agree with the reviewer, that tracking proteins by streptavidin imaging alone is problematic, because it cannot distinguish, which protein is biotinylated. We therefore used words like “likely” in the description of the data. However, we still think, it is a valid method, as long as it is confirmed by an orthogonal method. We have added this paragraph to the end of this chapter:

“Importantly, tracking of proteins by streptavidin imaging requires orthogonal controls, as the imaging alone does not provide information about the nature of the biotinylated proteins. These can be proximity ligation assay, mass spectrometry or specific tagging visualisation of protein suspects by fluorescent tags. Once these orthogonal controls are established for a specific tracking, streptavidin imaging is an easy and cheap and highly versatile method to monitor protein interactions in a specific setting.”

Reviewer #2 (Public Review):

Summary:

The authors noticed that there was an enhanced ability to detect nuclear pore proteins in trypanosomes using a streptavidin-biotin-based detection approach in comparison to conventional antibody-based detection, and this seemed particularly acute for phase-separated proteins. They explored this in detail for both standard imaging but also expansion microscopy and CLEM, testing resolution, signal strength, and sensitivity. An additional innovative approach exploits the proximity element of biotin labelling to identify where interacting proteins have been as well as where they are.

Strengths:

The data is high quality and convincing and will have obvious application, not just in the trypanosome field but also more broadly where proteins are tricky to detect or inaccessible due to phase separation (or some other steric limitations). It will be of wide utility and value in many cell biological studies and is timely due to the focus of interest on phase separation, CLEM, and expansion microscopy.

Thank you! We are glad you liked it.

Reviewer #3 (Public Review):

Summary:

The authors aimed to investigate the effectiveness of streptavidin imaging as an alternative to traditional antibody labeling for visualizing proteins within cellular contexts. They sought to address challenges associated with antibody accessibility and inconsistent localization by comparing the performance of streptavidin imaging with a

TurboID-HA tandem tag across various protein localization scenarios, including phase-separated regions. They aimed to assess the reliability, signal enhancement, and potential advantages of streptavidin imaging over antibody labeling techniques.

Overall, the study provides a convincing argument for the utility of streptavidin imaging in cellular protein visualization. By demonstrating the effectiveness of streptavidin imaging as an alternative to antibody labeling, the study offers a promising solution to issues of accessibility and localization variability. Furthermore, while streptavidin imaging shows significant advantages in signal enhancement and preservation of protein interactions, the authors must consider potential limitations and variations in its application. Factors such as the fact that tagging may sometimes impact protein function, background noise, non-specific binding, and the potential for off-target effects may impact the reliability and interpretation of results. Thus, careful validation and optimization of streptavidin imaging protocols are crucial to ensure reproducibility and accuracy across different experimental setups.

Strengths:

- *Streptavidin imaging utilizes multiple biotinylation sites on both the target protein and adjacent proteins, resulting in a substantial signal boost. This enhancement is particularly beneficial for several applications with diluted antigens, such as expansion microscopy or correlative light and electron microscopy.*
- *This biotinylation process enables the identification and characterization of interacting proteins, allowing for a comprehensive understanding of protein-protein interactions within cellular contexts.*

Weaknesses:

- *One of the key advantages of antibodies is that they label native, endogenous proteins, i.e. without introducing any genetic modifications or exogenously expressed proteins. This is a major difference from the approach in this manuscript, and it is surprising that this limitation is not really mentioned, let alone expanded upon, anywhere in the manuscript. Tagging proteins often impacts their function (if not their localization), and this is also not discussed.*
- *Given that BioID proximity labeling encompasses not only the protein of interest but also its entire interacting partner history, ensuring accurate localization of the protein of interest poses a challenge.*
- *The title of the publication suggests that this imaging technique is widely applicable. However, the authors did not show the ability to track the localization of several distinct proteins on the same sample, which could be an additional factor demonstrating the outperformance of streptavidin imaging compared with antibody labeling. Similarly, the work focuses only on small 2D samples. It would have been interesting to be able to compare this with 3D samples (e.g. cells encapsulated in an extracellular matrix) or to tissues.*

Recommendations for the authors:

To enhance the assessment from 'incomplete' to 'solid', the reviewers recommend that the following major issues be addressed:

Major issues:

- (1) *Anti-biotin antibodies in combination with TurboID labeling should be used to compare the signal/labelling penetrance to streptavidin results. That would show if*

elevated biotin deposition matters, or if it is really the smaller size, more fluors, and higher affinity of streptavidin that's making the difference.

We agree with the reviewer, that it would be very interesting to distinguish whether the increase in signal comes from the multiple biotinylation sites or from streptavidin being a very good binder, or perhaps from both, and whether the size matters (IgG versus streptavidin). However, this question is very hard to answer, as antibodies differ massively in their affinity to the antigen. Thus, even if anti-biotin would give a better signal than anti-HA, this could be either caused by the increase in antigen-number (more biotin than HA-tag) or by the better binding affinity, or by a combination, and it would not allow to truly answer the question. We have now tested anti-biotin antibodies, also in response to reviewer 1, and got a much poorer signal in comparison to anti-HA or streptavidin.

Please note that we made another attempt using nanobodies to target phase-separated proteins, to see, whether size matters (Fig. 2I). The nanobody did not stain Mex67 at the nuclear pores, but gave a weak nuclear signal for NOG1, which may suggest that the nanobody can slightly better penetrate than IgG, but it does not rule out that the nanobody simply binds with higher affinity. Reviewer 1 has suggested to use the Halo Tag with PEG-biotin: this would indeed allow to directly compare the streptavidin signal caused by the TurboID with a single biotin added by the Halo tag. Unfortunately, the PEG-biotin does not penetrate trypanosome cells. In conclusion, we are not aware of a method that would allow to establish why streptavidin but not IgGs can penetrate to phase separated areas. We therefore prefer to not overinterpret our data, but stick to what is supported by the data: “the inability to label phase-separated areas is not restricted to anti-HA but applies to other antibodies”.

(3) Figure 4 A-B. The validity of claiming the correct localization demonstrated by streptavidin imaging comes into question, especially when endogenous fluorescence, via the fusion protein, remains undetectable (as indicated by the yellow arrow at apex).

In this figure, the streptavidin imaging does NOT show the correct localisation of the bait protein, but it does show proteins from historic interactions that have a distinct localisation to the bait. We had therefore introduced this chapter with the paragraph below, to make sure, the reader is aware of the limitations (which we also see as an opportunity, if properly controlled):

“We found that in most cases, streptavidin labelling faithfully reflects the steady state localisation of a bait protein, e.g., the localisation resembles those observed with immunofluorescence or direct fluorescence imaging of GFP-fusion proteins. For certain bait proteins, this is not the case, for example, if the bait protein or its interactors have a dynamic localisation to distinct compartments, or if interactions are highly transient. It is thus essential to control streptavidin-based *de novo* localisation data by either antibody labelling (if possible) or by direct fluorescence of fusion-proteins for each new bait protein.”

In particular, on lines 450-460, there's a fundamental issue with the argument put forward here. It is not possible to formally know that the posterior labeling is ALPH1 vs. another part of the decapping complex that was associated with PABP2-Turbo, or if the higher detection capacity of the Turbo-biotin label is uncovering a novel localization of the PABP2. While it is likely that it is ALPH1, it is not possible to rule out other possibilities with this approach. These issues should be discussed here and more generally the possibility of off-target labeling with this approach should be addressed in the discussion.

We fully agree with the reviewer, that tracking proteins by streptavidin imaging alone is problematic, because it cannot distinguish, which protein is biotinylated. We therefore used

words like “likely” in the description of the data. However, we still think, it is a valid method, as long as it is back-uped by an orthogonal method. We have added this paragraph to the end of this chapter:

“Importantly, tracking of proteins by streptavidin imaging requires orthogonal controls, as the imaging alone does not provide information about the nature of the biotinylated proteins. These can be proximity ligation assay, mass spectrometry or specific tagging visualisation of protein suspects by fluorescent tags. Once these orthogonal controls are established for a specific tracking, streptavidin imaging is an easy and cheap and highly versatile method to monitor protein interactions in a specific setting.”

(4) More discussion and acknowledgment of the general limitations in using tagged proteins are needed to balance the manuscript, especially if the hope is to draw a comparison with antibody labeling, which works on endogenous proteins (not requiring a tag). For example: (a) tagging proteins requires genetic/molecular work ahead of time to engineer the constructs and/or cells if trying to tag endogenous proteins; (b) tagged proteins should technically be validated in rescue experiments to confirm the tag doesn't disrupt function in the cell/tissue/context of interest; and (c) exogenous tagged proteins compete with endogenous untagged proteins, which can complicate the interpretation of data.

We have added this paragraph to the first paragraph of the discussion part:

“Like many methods that are frequently used in cell- and molecular biology, streptavidin imaging is based on the expression of a genetically engineered fusion protein: it is essential to validate both, function and localisation of the TurboID-HA tagged protein by orthogonal methods. If the fusion protein is non-functional or mis-localised, tagging at the other end may help, but if not, this protein cannot be imaged by streptavidin imaging. Likewise, target organisms not amenable to genetic manipulation, or those with restricted genetic tools, are not or less suitable for this method.”

Also, we like to point out that for non-mainstream organisms like trypanosomes, antibodies are not commercially available and often genetic manipulation is more time-efficient and cheaper than the production of antiserum against the target protein.

Also, the introduction would ideally be more general in scope and introduce the pros and cons of antibody labeling vs biotin/streptavidin, which are mentioned briefly in the discussion. The fact that the biotin-streptavidin interaction is ~100-fold higher affinity than an IgG binding to its epitope is likely playing a key role in the results here. The difference in size between IgG and streptavidin, the likelihood that the tetrameric streptavidin carries more fluors than a IgG secondary, and the fact that biotin can likely diffuse into phase-separated environments should be clearly stated. The current introduction segues from a previous paper that a more general audience may not be familiar with.

We have now included this paragraph to the introduction:

“It remains unclear, why streptavidin was able to stain biotinylated proteins within these antibody inaccessible regions, but possible reasons are: (i) tetrameric streptavidin is smaller and more compact than IgGs (60 kDa versus a tandem of two IgGs, each with 150 kDa) (ii) the interaction between streptavidin and biotin is ~100 fold stronger than a typical interaction between antibody and antigen and (iii) streptavidin contains four fluorophores, in contrast to only one per secondary IgG.”

Minor issues:

The copy numbers of the HA and Ty1 epitope tags vary depending on the construct being used. For example, Ty1 is found as a single copy tag in the TurboID tag, but on the mNeonGreen tag there are 6 copies of the epitope. It makes it hard to know if differences in detection are due to variations in copies of the epitope tags. Line 372-374: can the authors explain why they chose to use nanobodies in this case? It would be great to show the innate mNeonGreen signal in 2K to compare to the Ty1 labeling. The presence of 6 copies of the Ty1 epitope could be essential to the labeling seen here.

We agree with the reviewer, that these data are a bit confusing. We have now removed Figure 3K, as it is the only construct with 6 Ty1 instead of one, and it does not add to the conclusions. (the mNeon signal is entirely in the nucleolus, as shown by TrypTag). We have also added an explanation why we used nanobodies (“The absence of a nanobody signal rules out that it’s simply the size of IgGs that prevents the staining of Mex67 at the nuclear pores, as nanobodies are smaller than (tetrameric) streptavidin”). However, as stated above, we prefer not to overinterpret the data, as signals from different antibodies/nanobodies – antigen combinations are not comparable. Important to us was to stress that the absence of signal in phase-separated areas is NOT restricted to the anti-HA antibody, which is clearly supported by the data.

What is the innate streptavidin background labeling look like in cells that are not carrying a TurboID fusion, from the native proteins that are biotinylated? That should be discussed.

We have now included the controls without the TurboID fusions for trypanosomes and HeLa cells: “Wild type cells of both Trypanosomes and human showed only a very low streptavidin signal, indicating that the signal from naturally biotinylated proteins is neglectable (Figure S8 in supplementary material).”

Line 328-331: This is likely to be dependent on whether or not the protein moves to different localizations within the cell.

True, we agree, and we have added this paragraph:

“The one exception are very motile proteins that produce a “biotinylation trail” distinct to the steady state localisation; these exceptions, and how they can be exploited to understand protein interactions, are discussed in chapter 4 below.”

Line 304-305: Does biotin supplementation not matter at all?

No, we never saw any increase in biotinylation when we added extra biotin to trypanosomes. The 0.8 μ M biotin concentration in the medium were sufficient.

Line 326-327: Was the addition of biotin checked for enhancement in the case of the mammalian NUP98? I would argue that there is a significant number of puncta in Figure 1D that are either green or magenta, not both. The amount of extranuclear puncta in the HA channel is also difficult to explain. Biotin supplementation to 500 μ M was used in mammalian TurboID experiments in the original Nature Biotech paper- perhaps nanomolar levels are too low.

We now tested HeLa cells with 500 μ M Biotin and saw an increase in signal, but also in background; due to the increased background we conclude that low biotin concentrations are more suitable. We have also repeated the experiment using 4HA tags instead of 1HA, and we found a minor improvement in the antibody signal for NUP88 (while the phase separated NUP54 was still not detectable). We have replaced the images in Figure 1D (NUP88) and also

in Figure 2F (NUP54) with improved images and using 4HA tags. However, we like to note that single nuclear pore resolution is beyond what can be expected of light microscopy.

Line 371: In 2I, I see a signal that looks like the nucleus, similar to the Ty1 labeling in 2G, so I don't think it's accurate to say that that Mex67 was "undetectable". Does the serum work for blotting?

Thank you, yes, “undetectable” was not the correct phrase here. Mex67 localises to the nuclear pores, to the nucleoplasm and to the nucleolus (GFP-tagging or streptavidin). Antibodies, either to the tag or to the endogenous proteins, fail to detect Mex67 at the nuclear pores and also don't show any particular enrichment in the nucleolus. They do, however, detect Mex67 in the (not-phase-separated) area of the nucleoplasm. We have changed the text to make this clearer. The Mex67 antiserum works well on a western blot (see for example: Pozzi, B., Naguleswaran, A., Florini, F., Rezaei, Z. & Roditi, I. The RNA export factor TbMex67 connects transcription and RNA export in *Trypanosoma brucei* and sets boundaries for RNA polymerase I. *Nucleic Acids Res.* **51**, 5177–5192 (2023))

Line 477: "lacked" should be "lagged".

Thank you, corrected.

Line 468-481: My previous argument holds here - how do you know that the difference in detection here is just a matter of much higher affinity/quantity of binding partner for the avidin?

See answer to the second point of (3), above.

483-491: Same issue - without certainty about what the biotin is on, this argument is difficult to make.

See answer to the second point of (3), above.

Line 530: "bone-fine" should be "bonafide"

Thank you, corrected.

Line 602: biotin/streptavidin labeling has been used for expansion microscopy previously (Sun, Nature Biotech 2021; PMID: 33288959).

Thank you, we had overlooked this! We have now included this reference and describe the differences to our approach clearer in the discussion part:

“Fluorescent streptavidin has been previously used in expansion microscopy to detect biotin residues in target proteins produced by click chemistry (Sun et al., 2021). However, to the best of our knowledge, this is the first report that employs fluorescent streptavidin as a signal enhancer in expansion microscopy and CLEM, by combining it with multiple biotinylation sites added by a biotin ligase. Importantly, for both CLEM and expansion, streptavidin imaging is the only alternative approach to immunofluorescence, as denaturing conditions associated with these methods rule out direct imaging of fluorescent tags.”

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